

Gregory R. Kronisch
Department of Biology and Ecology
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EDUCATION

University of Maine, Orono, Maine

- **Doctor of Philosophy**, Ecology and Environmental Science (June 2023 - Present)
 - Dissertation Topic: Movement and Diet of Maine's Landlocked Arctic Charr (*Salvelinus alpinus*) Populations in the Context of Climate Change
 - Supervisors: Michael T. Kinnison & Nathan B. Furey (University of New Hampshire)

State University of New York College of Environmental Science and Forestry, Syracuse, New York

- **Master of Science**, Fish and Wildlife Biology and Management (January 2017 - May 2019)
 - Thesis: Fish Assemblage Succession Within a Recovering Urban Lake
 - Supervisor: Neil H. Ringler
 - GPA 3.86
- **Bachelor of Science**, Aquatic and Fisheries Science, Cum Laude (August 2011 - May 2015)
 - Marine Sciences Minor, Honors Program
 - Supervisor: Kimberly Schulz
 - GPA 3.32

TEACHING EXPERIENCE

Graduate Teaching Assistant – Freshwater Fisheries Ecology and Management Practicum

(instructed by Dr. Stephen Coghlan) – University of Maine (September 2023) (*Junior level course. 30 students. Prepared and assisted with two day-length field trips, including operation of a motorboat and backpack electrofishing*)

Graduate Teaching Assistant – Ichthyology (instructed by Dr. Donald Stewart) - State University of New York College of Environmental Science and Forestry (January – May 2017)

(*Senior/graduate level course. 40 students. Prepared and instructed 2 lab sections including two day-length field trips; designed and graded multiple lab practicals; graded multiple full-length lab reports*)

Undergraduate Teaching Assistant – General Biology I (instructed by Dr. Gregory McGee) - State University of New York College of Environmental Science and Forestry (September – December 2013). Freshman level course. (*30 students. Prepared and instructed 1 lab section; graded all lab assignments*)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (GS-9 Equivalent) - University of Maine, Orono, Maine (June 2023 – Present)

- Evaluate structure of the Arctic Charr population of in Floods Pond, Otis, Maine
- Construct and maintain high precision acoustic telemetry array in Floods Pond
- Supervise multiple undergraduate technicians for fisheries field work and laboratory analyses
- Teaching assistant for Freshwater Fisheries Ecology and Management Practicum (September 2023)

Fish Biologist (GS-9) – United States Fish and Wildlife Service Lower Great Lakes Fish and Wildlife Conservation Office, Basom, New York (September 2019 – May 2023)

- Provide fishery technical assistance on areas under the jurisdiction of federal agencies and other cooperators, including the New York State Department of Environmental Conservation, US Geological Survey, and St Regis Mohawk Tribe Environmental Division
- Assist interagency cooperative projects to restore Lake Sturgeon, Lake Trout, and coregonine populations in Lake Ontario, Lake Erie, Niagara River, Genesee River, and St. Lawrence River
- Deploy and maintain multiple acoustic telemetry arrays within the Great Lakes Acoustic Telemetry Observation System (GLATOS) and manage associated Microsoft Access database
- Develop statistical code in R for analysis of biotelemetry data and overall data quality control

Associate Environmental Scientist (GS-5 Equivalent) - Parsons Corporation, Syracuse, New York (June - July 2019)

- Administered chemical applications to Onondaga Lake benthos to mitigate mercury methylation
- Assisted with rigging and installation of industrial piping

Graduate Research Project Assistant (GS-7 Equivalent) - State University of New York College of Environmental Science and Forestry, Syracuse, New York (January 2017 - May 2019)

- Led 6-person field crew, including scheduling, training, and technical support
- Collected and identified fish species with trap net, gill net, beach seine, and boat electrofishing
- Conducted lake-wide macrophyte and Centrarchid nesting surveys
- Calculated estimates of Largemouth Bass population size from mark-recapture electrofishing surveys
- Lake Sturgeon mark-recapture via PIT and Carlin tags
- Prepared otoliths and scales of multiple sportfish species and aged by cross-section or
- Extensive data wrangling and management in R and MS Excel
- Teaching Assistant for Ichthyology course (January - May 2017)

Fisheries Research Technician (Contractor; GS-5 Equivalent) - U.S. Geological Survey (Tunison Laboratory of Aquatic Science), Cortland, New York (June 2015 – January 2017)

- Fish collection/identification, site characterization, and data management/analysis, GIS
- Evaluation of fish and Unionid mussel assemblages above and below Hogansburg Hydroelectric Dam on the St. Regis River prior to removal
- Coordinated data exchange between federal and state agencies
- Lake Sturgeon population assessment in New York tributaries to the St. Lawrence River

- Experimental and field eDNA sample collection in the St. Regis River, NY
- Aquaculturist for Atlantic Salmon (*Salmo salar*), Lake Herring (*Coregonus artedii*), and Bloater (*Coregonus hoyi*)

Undergraduate Researcher - SUNY College of Environmental Science and Forestry, Syracuse, New York

- Honor's Research Quantifying Decomposition of Allochthonous Matter in Forest Pools of Central New York (September 2014 - May 2015)
 - o Role of microbial decomposition on water chemistry and pool ecosystem
- Teaching Assistant - General Biology I Laboratory, SUNY ESF (Fall 2013)
 - o Instructed 3 hour weekly general biology lab along with a graduate TA
- Lab Research for Dr. Kim Schulz (October 2012 - May 2013)
 - o Benthic invertebrate sorting and macrophyte sample measurements

Student Intern - John Carroll University, Cleveland, Ohio (May - August 2014)

- Field research studying evolutionary trends in color morphology of the eastern red-backed salamander, *Plethodon cinereus*, across the Great Lakes Region
- Processed specimen photographs with ImageJ to evaluate morphology

SKILLS

- Computer: R Statistical Software, Microsoft Office, Microsoft Access, Esri ArcGIS, ImageJ
- Outdoor: Knot tying, Map and Compass, SCUBA/Snorkeling
- Technical: Trailering/Operating Motor Boats, Standard Vehicle Operation, Motorboat Operators Certification Course
- Field Sampling: Fish/Amphibian/Macrophyte/Invertebrate Identification, Li-COR and YSI Meter Operation/Maintenance, Boat and Backpack Electrofishing, Seine/Gill/Trap/Plankton/Trawl netting, eDNA Collection
- Laboratory Sampling: Basic Phytoplankton/Zooplankton identification, Aquatic Invertebrate Identification, Spectrophotometry, Acid Washing, Chromatography

PRESENTATIONS

Kronisch, GR, MT Kinnison, CA Murphy, GT Schumacher, BF Erdman, T Banister and NB Furey. 2025. Seasonal Habitat Associations of Lacustrine Arctic Charr (*Salvelinus alpinus*). Atlantic International Chapter of the American Fisheries Society Annual Meeting, Lac-Delage, Quebec. (Oral)

Schumacher, GT, EB Peebles, MT Kinnison, NB Furey, **GR Kronisch**, and CA Murphy. 2025. Navigating a food web from planktivory to piscivory: variable trophic ontogeny in congeneric fish predators. Society for Freshwater Science Annual Meeting. San Juan, Puerto Rico. (Oral)

Kronisch, GR, MT Kinnison, CA Murphy, GT Schumacher, T Banister and NB Furey. 2025. Fine-

scale Habitat Use by Lacustrine Arctic Charr (*Salvelinus alpinus*) in a Maine, USA Lake. Maine Cooperative Fish and Wildlife Research Unit Annual Meeting. Orono, Maine. (Oral)

Kronisch, GR, MT Kinnison, CA Murphy, GT Schumacher, T Banister and NB Furey. 2024. Seasonal Movement Ecology of Landlocked Arctic Charr (*Salvelinus alpinus*) in a Maine Reservoir. Atlantic International Chapter of the American Fisheries Society Annual Meeting, Sunapee, New Hampshire. (Poster)

Kronisch, GR, MT Kinnison, CA Murphy, GT Schumacher, T Banister and NB Furey. 2024. Fine-scale Habitat Use by Landlocked Arctic Charr (*Salvelinus alpinus*) in a Maine, USA Lake. Ocean Tracking Network Symposium, Halifax, Nova Scotia. (Oral)

Schumacher, GT, CA Murphy, NB Furey, MT Kinnison, **GR Kronisch**, B Erdmann, and EB Peebles. 2024. Trophic ontogeny of an elusive fish through eye lens stable isotope analysis. 154th Annual Meeting of the American Fisheries Society. Honolulu, Hawai'i. (Poster)

Morton, K, D Gorsky, J Sweka, AT Fisk, TB Johnson, J Johnson, C Farrell, **G Kronisch**, L Paulic, S Ivanova, B Weidel, D Mason, A Gatch, R Walquist, L Le Tarte, G Hoffman, M Nguyen. Post-stocking Mortality and Behavior of Age-1 Lake Trout in Lake Ontario. 2024. New York Chapter of the American Fisheries Society Annual Meeting. Cooperstown, New York. (Oral)

Kronisch, G, and D Gorsky. Categorizing Behaviors of Lake Sturgeon in the Niagara River. 2023. New York Chapter of the American Fisheries Society Annual Meeting. Stony Brook, New York. (Oral)

Kronisch, G, D Gorsky, Z Biesinger. Long-term and Notable Movement Patterns of Lake Sturgeon (*Acipenser fulvescens*) in the Lake Ontario basin. 2022. Joint Aquatic Sciences Meeting. Grand Rapids, Michigan. (Oral)

Johnson, RJ, D Gorsky, C Castiglione, **G Kronisch**, A Gatch, Z Biesinger. Habitat Characterization for Juvenile Lake Sturgeon (*Acipenser fulvescens*) in the Upper Niagara River. 2022. Joint Aquatic Sciences Meeting. Grand Rapids, Michigan. (Poster)

Kronisch, G, D Gorsky, Z Biesinger. Long-term and Notable Movement Patterns of Lake Sturgeon (*Acipenser fulvescens*) in the Lake Ontario basin. 2022. New York Chapter of the American Fisheries Society Annual Meeting. Virtual. (Oral)

Gatch, A, D Gorsky, Z Biesinger, **G Kronisch**, C Hand. Continued use of a 40-year-old constructed spawning reef by Lake Trout. 2022. New York Chapter of the American Fisheries Society Annual Meeting. Virtual. (Oral)

Johnson, RJ, D Gorsky, C Castiglione, **G Kronisch**, A Gatch, Z Biesinger. Habitat Characterization for Juvenile Lake Sturgeon (*Acipenser fulvescens*) in the Upper Niagara River. 2022. New York Chapter of the American Fisheries Society Annual Meeting. Virtual. (Poster)

Kronisch, G, D Gorsky, Z Biesinger, D Dittman, M Chalupnicki, R Klindt, T Brooking, JR Jackson, T

Van De Valk, S Schlueter, N Ringler, E Zollweg-Horan. Early evaluation of lake-wide lake sturgeon (*Acipenser fulvescens*) movements in the context of Lake Ontario management units. 2020. New York Chapter of the American Fisheries Society Annual Meeting, Lake Placid, New York. (Poster)

Kronisch, GR and NH Ringler. Evaluation of Fish Assemblage Response to the Remediation of an Urban Lake. 2019. New York Chapter and Northeast Division of the American Fisheries Society Annual Meeting, Poughkeepsie, New York. (Oral)

Limburg, K, D Breitburg, J Cramer, SS Ekoh, A Gårdmark, Y Heimbrand, **G Kronisch**, J Le, LA Levin, LR McCormick, S McNulty, A Orio, MA Samson, S Shatto and KM Smith. 2018. Valuing Ecosystem Services at Risk from Deoxygenation of Oceans, Estuaries, and Coastal Seas. American Fisheries Society 2018 National Meeting, Atlantic City, New Jersey. (Oral)

Kronisch, GR, DJ Stewart, and NH Ringler. Longitudinal Changes of Fish Assemblages in Nine Mile Creek, Tributary to a Recovering Urban Lake. 2018. New York Chapter of the American Fisheries Society Annual Meeting, Cooperstown, New York. (Poster)

Kronisch, GR, JA DiRado, JH Johnson, and MA Chalupnicki. Re-establishing Atlantic Salmon into Lake Ontario via the Salmon River, New York. 2017. New York Chapter of the American Fisheries Society Annual Meeting, Buffalo, New York. (Poster)

Kronisch, GR, KS Hanak, and JE McKenna. Young of Year Habitat Preference in St. Lawrence River Tributaries. 2016. New York Chapter of the American Fisheries Society Annual Meeting, Cooperstown, New York. (Poster)

PUBLICATIONS

Schumacher, GT, EB Peebles, NB Furey, MT Kinnison, **GR Kronisch**, and CA Murphy. Accepted. Eye lens stable isotope analysis reveals variable trophic ontogeny among populations of freshwater fishes. *PLOS One*. DOI Pending.